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ResNet-50 Transfer Learning Model for Diabetic Foot Ulcer Detection Using Thermal Images

1st Aditya Kumar
Chitkara University Institute of Engineering & Technology
Chitkara University
Rajpur, Punjab, India
aditya.3891@chitkara.edu.in

2nd Leema Nelson
Chitkara University Institute of Engineering & Technology
Chitkara University
Rajpur, Punjab, India
leema.nelson@chitkara.edu.in

3rd Sarabjeet Singh
Chandigarh College of Engineering and Technology
Chandigarh, India
ssingh@ccet.ac.in

***Abstract*— Diabetic Foot Ulcer (DFU) is a major global healthcare challenge, entailing substantial costs and increased mortality rates. The aim of this work is to develop a ResNet-50 transfer learning model for DFU detection using thermal images. The ResNet-50 transfer learning model is fine-tuned to adapt its pre-trained features to DFU detection, yielding improved accuracy and dataset generalisation. The developed model achieves 90.52% accuracy, 92.06% precision, 96.97% AUC, and 92.06% recall, showing significant performance gains. Furthermore, a 7% increase in accuracy is observed in comparison to the previous state-of-the-art. This work presents substantial enhancements in DFU detection accuracy, highlighting the potential of deep learning and data augmentation in medical image analysis. The developed model's improved performance underscores its relevance for clinical applications, suggesting its potential to enhance patient care and outcomes.**

Keywords*— *Diabetic foot ulcer, DFU, Adam, deep learning, CNN, ResNet-50, Thermal Images.

I. Introduction

Diabetic Foot Ulcers (DFUs) pose a critical global healthcare concern, giving rise to substantial healthcare expenditures and heightened mortality rates [1]. DFUs represent foot injuries that occur in cases of severe diabetes. According to multiple reports, the worldwide count of individuals diagnosed with diabetes was 151 million in the year 2000. However, this number witnessed a significant surge, exceeding 422 million by 2014, with projections indicating a further rise beyond 537 million by 2021. Notably, over the span from 2000 to 2021, there was a marked 10.5% increase in the prevalence of diabetes among individuals aged 18 and older. There will be 630 million diabetics worldwide by the end of 2035, according to studies. Around 80% of these patients are situated in underdeveloped nations characterised by inadequate healthcare infrastructure and limited awareness of patient health matters. [2]. Diabetic feet affect 15–25% of diabetic patients, and they may go through the last stage of foot ulcers. In some cases, the lower limbs need to be removed, followed by hospitalisation, and lastly, with poor medical treatment, the patient dies [3,4]. The possibility of foot or limb amputation arises due to the infection stemming from diabetic foot ulcers [5]. Approximately 1 million diabetics lose a section of their legs each year due to incorrect DFU diagnosis and treatment [6]. As mentioned previously, to avert future implications, diabetic patients with high-risk foot conditions necessitate consistent medical check-ups, ongoing investment in expensive medications, and meticulous personal care. Consequently, this scenario puts a significant economic load on individuals and their families, a challenge particularly pronounced in less developed countries. In such contexts, the cost associated with handling this ailment can reach up to 5.7 times their yearly income. [7]. Lower survival rates are shown in patients who have had limbs removed. Along with living standards and livelihood, social participation is also affected [8]. It has been noted that approximately every 20 seconds, surgical intervention is performed on a patient with a diabetic foot condition. To establish a precise diagnosis, medical practitioners need to analyse medical data extensively. Conventional diagnostic methods require significant human effort and are susceptible to error. The use of computer-assisted diagnostic approaches not only decreases costs but also enhances accuracy. Contemporary advancements in mobile and wearable health devices have played a crucial role in managing diabetes and its associated complications. These devices aid in prolonging remission, and they also contribute to enhancing the quality of life for patients. They achieve this by monitoring and regulating detrimental factors such as foot pressure and inflammation [1]. The most basic monitoring approach, visual inspection, comes with inherent limitations. This is particularly problematic for individuals dealing with obesity or visual impairments as they might struggle to accurately detect subtle changes. Encouragingly, recent research underscores the promise of a home-based temperature monitoring system, showcasing an impressive 97% accuracy in identifying DFUs at their early stages [9]. This underscores the significance of accessible and efficient monitoring methods [10].

Individuals who receive consistent temperature monitoring of their feet encounter a reduced probability of developing complications related to foot health. The assessment of Diabetic Foot Ulcers (DFUs) within contemporary clinical protocols encompasses a range of critical duties, encompassing prompt detection and continuous tracking of advancements. This encompasses a sequence of labour-intensive procedures intended to guarantee tailored therapeutic interventions and meticulous attention for every instance of DFU. Initially, a thorough review of the patient's medical background is conducted, succeeded by a comprehensive analysis performed by specialists with expertise in diabetic foot conditions [11]. This study introduces a novel approach for automated Diabetic Foot Ulcer (DFU) detection through the utilisation of thermal images, leveraging the power of convolutional

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neural networks (CNNs). Specifically, the ResNet-50 architecture is utilised, a deep CNN renowned for its exceptional feature extraction capabilities. CNNs are crafted to autonomously and flexibly understand layered patterns of attributes from input information. This attribute equips them with remarkable efficacy for undertakings such as object detection and image classification [14]. The significance of this work lies in the fine-tuning process applied to the ResNet-50 model, which aims to adapt the pre-trained network to the task of DFU detection. By fine-tuning, the refinement of the model's parameters to achieve higher accuracy and better generalisation of the specific dataset.

II. Literature Review

Goyal et al. (2020) explore computerised methods for analysing DFUs, focused on infection and ischaemia recognition. They introduced a significant dataset with verified infection and ischaemia labels, using a handcrafted machine-learning approach and an Ensemble CNN model. This method showcased better ischaemia classification compared to infection, achieving notable accuracy rates of 90% for ischaemia and 73% for infection. The study advances the use of computerised techniques for assessing DFUs [1].

Thotad et al. (2023) conducted research aimed at enhancing the detection of DFUs through the utilisation of deep neural networks. Their investigation employed a comprehensive dataset comprising 844 images of feet, encompassing both healthy and ulcer-affected cases, and the study employed EfficientNet to achieve early detection. This approach outperformed prior models (AlexNet, GoogleNet, VGG16, VGG19) with exceptional accuracy (98.97%), f1-score (98%), recall (98%), and precision (99%). The study underscores the capacity of deep learning to improve the prognosis and management of diabetic ulcers [2]. Goyal et al. tackled the issue of identifying diabetic foot ulcers (DFUs) through an inventive machine-learning approach. Their research collected an extensive dataset containing images of feet afflicted with DFUs as well as images of healthy skin. They performed binary classification to distinguish normal skin from DFUs. The study introduced DFUNet, a novel CNN architecture designed to enhance the extraction of distinctive features between unaffected skin and DFUs. Through a 10-fold cross-validation approach, the proposed model demonstrated remarkable performance with an AUC score of 0.961, outperforming traditional deep learning and machine learning classifiers. This pioneering work offers a highly sensitive DFU detection solution, potentially transforming diabetic foot care by enabling remote, cost-effective, and user-friendly healthcare [7].

Ahsan et al. (2023) conducted research aimed at improving the classification of DFUs using deep neural networks. By utilising a dataset containing 844 images of feet encompassing both healthy and those affected by ulcers, the study employed EfficientNet to attain early detection. This method exhibited superior performance compared to earlier models such as AlexNet, GoogleNet, VGG16, and VGG19, achieving remarkable levels of accuracy (99%), f1-score (98%), recall (98%), and precision (99%) [10].

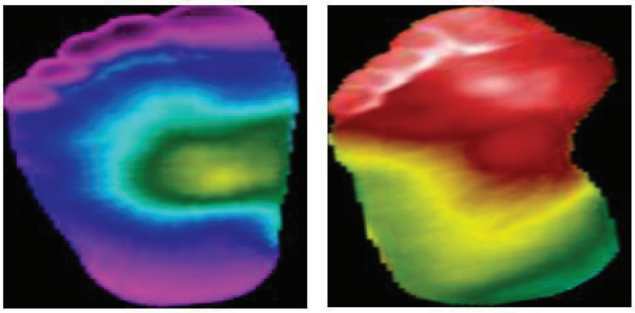
Munadi et al. (2022) proposed an innovative framework for categorising DFUs employing thermal imaging and deep neural networks. The study utilised convolutional neural network models, including MobileNetV2 and ShuffleNet, as baseline classifiers, training them on datasets comprising plantar thermograms. Through a decision fusion technique, the classification outcomes from these models were combined, resulting in a 100% accuracy rate in distinguishing positive and negative DFU cases. This approach demonstrated a 3.4% accuracy improvement over the baseline classifiers and showcased superior performance compared to state-of-the-art deep learning and conventional machine learning-based classifiers for image classification [12].

III. Methodology

This work centres on the detection of foot ulcers, utilising the ResNet-50 model within the context of deep learning. The dataset is divided into two parts: Thermal images of Healthy feet and DFUs. Data processing techniques and utilising the ResNet-50 transfer learning model within the context of deep learning and get the results. A systematically developed procedure, incorporating essential components, is adopted to successfully attain this objective.

A. Database Description

This work obtained DFUs thermal images from the "Thermography images of diabetic foot" dataset obtained from the Kaggle repository. The dataset comprises thermal foot images, which are categorised into two groups: the control group and the diabetic mellitus (DM) Group. The control group consists of 894 images, and the DM group includes 923 images. The images of the Control Group and DM Group are shown in Fig. 1(a) and 1(b).



(a) (b)

Fig. 1. Thermal Images of (a) Healthy Foot and (b) Diabetic Mellitus Foot.

B. Data Augmentation

In this work, data augmentation techniques are employed to improve the robustness and generalisation of the deep learning model. These techniques are applied to the thermography images of diabetic feet to create a more diverse training dataset. The implemented data augmentation techniques include:

a. *Rotation:* To simulate different foot orientations, images are subjected to random rotation by an angle within a predefined range.
b. *Shear:* Shearing transformations are applied to the images to introduce perspective distortion, mimicking variations in camera angles.
c. *Zoom:* Random zooming is performed on the images to simulate varying distances between the camera and the subject.

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d. *Horizontal Flip:* Images are horizontally flipped with a certain probability of capturing the mirror reflections of foot patterns.

e. *Width and Height Shift:* Shifts in the width and height dimensions of the images are applied to simulate variations in the foot's position.

These data augmentation techniques are employed using the Image Data Generator class from the TensorFlow library, enhancing the diversity of the training dataset.

C. Model Architecture

This work's primary deep learning architecture is based on a modified version of the ResNet-50 model, as shown in Fig.2. The model is initialised with pre-trained weights from the ImageNet dataset. The architecture consisted of the following layers:

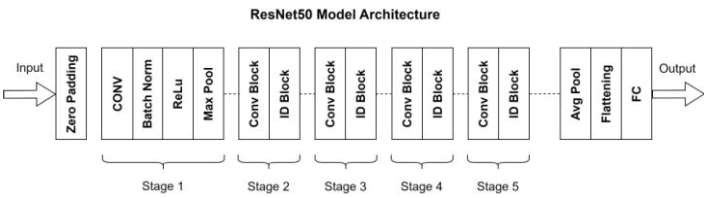

Fig. 2. ResNet-50 Model Architecture

a. *ResNet-50 Base:* A pre-trained ResNet-50 model is used as the base architecture to capture high-level features from the input images [13]. The layers of the base model are kept frozen during training.

b. *Global Average Pooling:* To decrease the spatial dimensions of the feature maps and retain crucial features, a global average pooling layer is introduced.

c. *Dense Layers:* Fully connected dense layers are added for feature transformation. A dense layer with ReLU activation and batch normalisation is employed.

d. *Model Training:* The Resnet-50 model is developed using the training dataset. The training dataset comprises a total of 1444 images, with 744 of them belonging to the control group and 700 belonging to the DM Group, taken from 1444 images. The model is put together using the binary cross-entropy loss method and improved with the Adam optimiser. Also, measurement tools like accuracy, area under the curve (AUC), precision, and recall are used while teaching. To ensure good progress, a Model Checkpoint callback is used to save the model with the best validation accuracy during the training.

e. *Evaluation*: The Resnet-50 model is evaluated using the validation dataset consisting of 422 images, of which 150 images belonged to the Control Group, and 223 images belonged to the DM Group, and the performance is assessed on both the validation and training datasets using metrics like accuracy, AUC, precision, and recall. These metrics provide important information on how well the model can categorise DFUs.

f. *Gaussian Noise:* To improve the model's robustness and minimise overfitting, the model incorporates Gaussian noise.

g. *Dropout:* In order to decrease overfitting, a dropout layer is added during training by randomly deactivating neurons.

h. *Output Layer:* The final dense layer used a sigmoid activation function to output a binary detection indicating the presence or absence of diabetic foot symptoms.

IV. PROPOSED WORK

The primary purpose of this work is to increase the accuracy of Diabetic Foot Ulcer detection using a ResNet-50 pre-trained model sourced from the Kaggle repository. The aim is to fine-tune the model and optimise its performance using the Adam optimisation algorithm to maximise its performance.

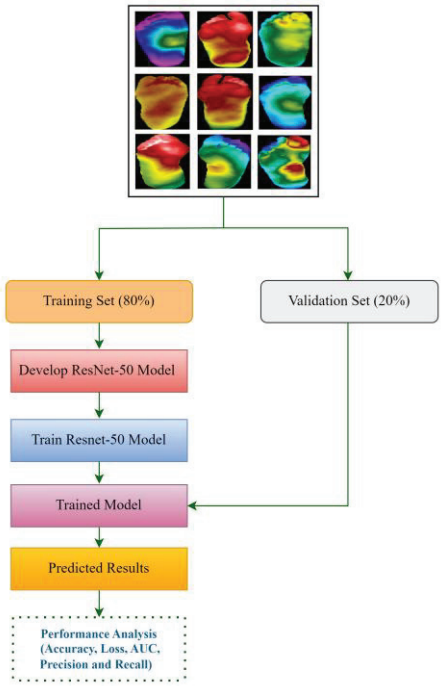

Fig. 3. Proposed Work Flow

The approach involves the utilisation of the "Thermography images of the diabetic foot" dataset [15], as shown in Fig. 3. This dataset is divided into two subsets: an 80% training set and a 20% validation set. The training set comprises 1,493 images, while the validation set consists of 373 images. The process begins with the development of the ResNet-50 model, which is then trained using the designated training dataset. To enhance the model's robustness, data augmentation techniques are employed on the training dataset. Following this, the model is refined through fine-tuning using the validation dataset The outcomes are then measured in terms of accuracy, loss, Area Under the Curve (AUC), precision, and recall, providing comprehensive insight into the model's predictive performance.

V. RESULTS AND DISCUSSION

In this section, we will showcase the outcomes of this work and explains the implications of each one. Firstly,

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summarise the model training process and subsequently discuss the achieved results.

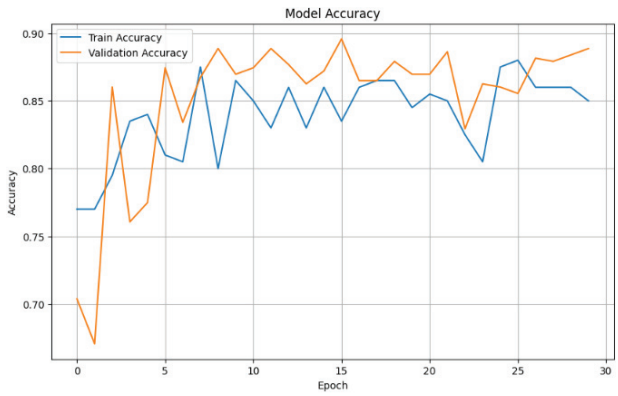

Fig. 4. Training and Validation Accuracy

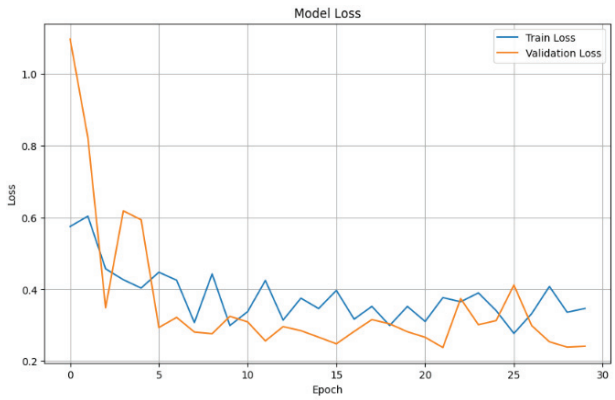

Fig. 5. Training and Validation Loss

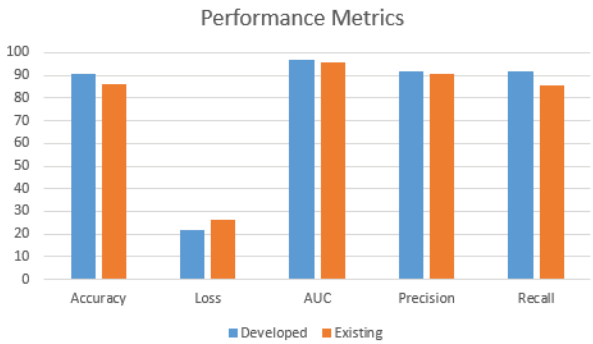

Fig. 6. Performance Metrics

A. *Experimental Setup:*

a) The experiments are conducted on the "Thermography images of diabetic foot" dataset, which consists of thermal images of both healthy and ulcerated feet. The developed model is implemented using the ResNet-50 architecture with additional layers for classification.

B. *Performance Metrics:*

b) The model's effectiveness is assessed using measures like accuracy, recall, precision, and the area under the receiver operating characteristic curve (AUC). These measures give a thorough evaluation of how well the model can accurately sort different cases of DFUs. The results of the developed model were compared with the existing method, which represents the best-performing approach prior to work.

C. *Results and Interpretation:*

The developed RedNet-50 pre-trained model achieved a significant improvement in all performance metrics compared to the existing method. The accuracy increased by 7%, indicating the model's ability to correctly classify a higher proportion of both healthy and ulcerated feet. The AUC score, which measures the model's overall discrimination ability, improved by 6%. The higher precision value indicates that the model has a reduced false positive rate, reducing the misclassification of healthy feet as ulcerated. Moreover, the higher recall value indicates that the model better identifies actual DFUs, minimising false negatives. The training and validation progress is shown in Fig.2, which shows the improvement in accuracy during the training process. Additionally, Fig. 3 shows the training and validation loss over epochs, providing insights into the model's convergence and generalisation capabilities.

The observed improvements can be attributed to several factors. The model benefits from using more advanced architecture, incorporating deep learning techniques and data augmentation. The extra layers and methods, like batch normalisation and dropout, help the model learn more complex patterns and decrease overfitting.

Furthermore, the augmentation of the training data helps the model generalise better to previously unseen images. Including more diverse angles, rotations, and shifts aids the model in capturing variations present in real-world scenarios. The comparison between our developed work and existing work is shown in Fig. 6 and Table 1.

TABLE I. PERFORMANCE PARAMETERS SUMMARY.

Model	Accuracy (%)	Loss (%)	AUC (%)	Precision (%)	Recall (%)
Developed	90.52	21.62	96.97	92.06	92.06
Existing	86.02	26.46	95.62	90.72	85.32

The results show that utilising deep learning techniques can successfully categorise DFUs through thermal images. This technology could have a significant impact on early detection and intervention for diabetic patients, ultimately reducing the risk of complications.

VI. CONCLUSION

The developed approach showcased substantial improvements in DFUs classification accuracy in contrast to the current state-of-the-art technique. In this work, the

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ResNet-50 model was employed for precise DFU classification based on thermal images. The results demonstrated a notable enhancement in the classification of DFUs with 90.5% accuracy, 97% AUC, 21.6% Loss, Recall 92% and 92.6% precision on the dataset compared to the previous approach, as shown in Fig. 6. The training and validation progress indicated notable enhancement, as shown in Fig. 4, while Fig. 5 shown the corresponding reduction in training and validation loss. This highlights the potential of deep learning techniques and data augmentation in medical image analysis. The enhanced performance of the model underscores its relevance in the clinical setting and its potential to contribute to improved patient care and outcomes. Future work could focus on exploring ensemble methods to further enhance classification performance. Additionally, investigating domain-specific features extracted from thermal images could provide deeper insights into the characteristics of DFUs.

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